Permute row and columns of matrices

Consider a matrix $A \in \mathbf{R}^{n\times n}$ and given a linear index $p = (1, 2, \ldots, n)$. Suppose $n = 4$ and we have a sequence of two indices as $(2, 3), (4, 1), (3, 1)$, our goal is to perform two tasks and return two answers:

1. Perform row interchanges on A according to the two indices in each sequence. For example, we perform $R_2 \Leftrightarrow R_3$ on A to get A_1 and then perform $R_4 \Leftrightarrow R_1$ on A_1 to get A_2, and last perform $R_3 \Leftrightarrow R_1$ on A_2 to get A_3. Then you return A_3 as the answer.
2. Perform column interchanges on A according to the two indices in each sequence. In this case, first you swap columns 2 and 3, to get C_1. Next, you swap columns 4 and 1 of C_1 to get C_2. Last, you swap columns 3 and 1 on C_2 to get C_3. Then you return C_3 as the answer.

In general, a sequence of two indices are provided as $\mathcal{I} = \{(i_1, j_1), (i_2, j_2), \ldots, (i_K, j_K)\}$. The index set has K members, meaning that you perform K steps of row/column swapping. You can see that performing row/column interchanges on A just involves how to keep track of the linear index p according to the swapping information. Back to the example, we see that the linear index p evolves as

$$\text{step } 0{:}p = (1, 2, 3, 4),\ \ \text{step } 1{:}p = (1, 3, 2, 4),\ \ \text{step } 2{:}p = (4, 3, 2, 1),\ \ \text{step } 3{:}p = (2, 3, 4, 1)$$

The final row-permuted A is the matrix that has row changed according to $p = (2, 3, 4, 1)$ which gives the new order of rows in the result. Specifically, the rows of permuted A start from row 2,3,4, and then 1 from A. Similarly, the final column-permuted A has the rows starting from 2,3,4, and 1 from the columns of A.

Task

Given the index set $\mathcal{I}$, and a square matrix A of size n, return

1. The final linear index p corresponding to the final result of performing index swapping given by $\mathcal{I}$.
2. The row-permuted A according to swapping information in $\mathcal{I}$.
3. The column-permuted A according to swapping information in $\mathcal{I}$.

Write a function named `perm_mat(A,I)`.

Input Format

- $A \in \mathbf{R}^{n\times n}$ where $5 \le n \le 1000$ as an ND array.
- I as a 2D array of size $K \times 2$. Each row contains an index (i, j) for $i \neq j$. The values in I are positive integers ranging from 1 to n

Output Format

- p as a 1D array specifying the resulting linear index according to the result of permutation. Its values are distinct positive integers ranging from 1 to n.
- $A_p \in \mathbf{R}^{n\times n}$ is the result of row permutation on A.
- $C_p \in \mathbf{R}^{n\times n}$ is the result of column permutation on A.

Sample Input

```
Original index [1 2 3 4]
I is
 [[2 1]
 [1 3]
 [4 2]]
```

```
A is
 [[1 2 9 9]
 [9 4 5 8]
 [7 7 3 4]
 [3 4 5 5]]
```

Sample Output

```
Resulting index [3 4 2 1]

Row-permuted A is
 [[7 7 3 4]
 [3 4 5 5]
 [9 4 5 8]
 [1 2 9 9]]
Column-permuted A is
 [[9 9 2 1]
 [5 8 4 9]
 [3 4 7 7]
 [5 5 4 3]]
```

Implementation

Goal: Fill in the following function:

```
def perm_mat(A,I):

  return p,Ap,Cp
```

Make sure you import any library you use to solve the problem.